May 2024

IB MAAHL P1 TZ1 Student A

1. mean $= 3$

(a) $$\frac{p+2q+3(4)+4(2)+6(3)}{16} = 3.$$

$p+2q+38=48$

$\Rightarrow p+2q=10 \quad — (1)$

$p+q+4+2+3=16$

$p+q=7 \quad —(2)$

$(1)-(2): \quad q=3$

$\Rightarrow \quad p=4.$

(b) Mean final score $= 10(3)$

$= 30.$

2. $\log_{10} a = \frac{1}{3}, \quad a > 0$

(a) $\log_{10}\left(\frac{1}{a}\right) = \log_{10}(1) - \log_{10} a$

$= -\frac{1}{3}.$

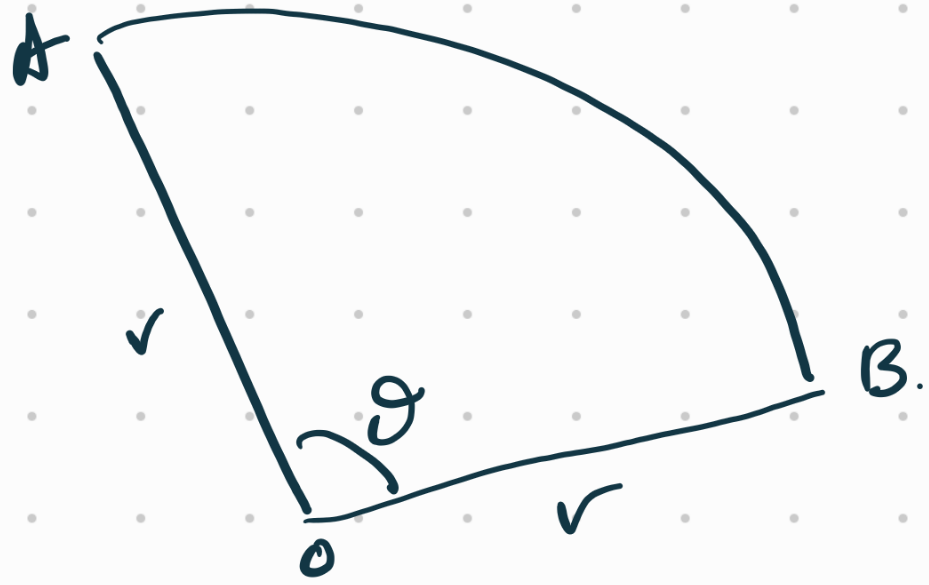
(b) $\log_{1000} a = \dfrac{\log_{10} a}{\log_{10} 1000}$

$= \dfrac{\frac{1}{3}}{\log_{10} 10^3}$

$= \dfrac{\frac{1}{3}}{3}$

$= \frac{1}{9}$

3.



(a) Perimeter $= 2r + r\theta = 10$ — (1)

Area $= \frac{1}{2}r^2\theta = 6.25$

$\Rightarrow r^2\theta = 12.5$

$\Rightarrow \theta = \frac{12.5}{r^2}$ — (2)

sub. (2) into (1).

$2r + r\left(\frac{12.5}{r^2}\right) = 10$

$2r^2 + 12.5 = 10r$.

$4r^2 - 20r + 25 = 0$ (shown)

(b) $4r^2 - 20r + 25 = 0$

$(2r - 5)(2r - 5) = 0$

$r = \frac{5}{2}.$

$\theta = \frac{12.5}{r^2} = \frac{12.5}{(\frac{5}{2})^2}$

$= \frac{25}{2} \times \frac{4}{25}$

$= 2$ rad.

4. $f(x) = \cos x$, $g(x) = \sin 2x$

$0 \leq x \leq \pi$.

(a) $\cos x = \sin 2x$.

$\Rightarrow \cos x = 2\sin x \cos x$

$\cos x (1 - 2\sin x) = 0$

$\cos x = 0$ (Rejected as $\cos x \neq 0$) or $\sin x = \frac{1}{2}$.

$x = \frac{\pi}{6}$ or $\pi - \frac{\pi}{6}$

$= \frac{5\pi}{6}$.

x-coordinate of $A = \frac{\pi}{6}$

x-coordinate of $C = \frac{5\pi}{6}$

(b) Area of R

$$= \int_{\frac{\pi}{2}}^{5\pi/6} (\cos x - \sin 2x)\, dx$$

$$= \left[\sin x + \frac{\cos 2x}{2} \right]_{\pi/2}^{5\pi/6}$$

$$= \left[\sin\left(\frac{5\pi}{6}\right) + \frac{1}{2}\cos\left(\frac{5\pi}{3}\right)\right] - \left[\sin\frac{\pi}{2} + \frac{1}{2}\cos\pi\right]$$

$$= \left[\frac{1}{2} + \frac{1}{2}\left(\frac{1}{2}\right)\right] - \left[1 - \frac{1}{2}\right]$$

$$= \frac{3}{4} - \frac{1}{2}$$

$$= \frac{1}{4} \text{ units}^2.$$

5. G.P. $u_1 = 1$, $r = 10$.

(a) $S_n = \frac{1(10^n - 1)}{10 - 1} = \frac{10^n - 1}{9}$

$a = 10$, $b = 9$

(b) $S_1 + S_2 + S_3 + \dots + S_n$

$= \frac{10^1 - 1}{9} + \frac{10^2 - 1}{9} + \frac{10^3 - 1}{9} + \dots + \frac{10^n - 1}{9}$.

$= \frac{1}{9}\left[10^1 + 10^2 + 10^3 + \dots + 10^n - (1 + 1 + 1 + \dots + 1)\right]$

$= \frac{1}{9}\left[\frac{10(10^n - 1)}{10 - 1} - n\right]$

$= \frac{1}{9}\left[\frac{10(10^n - 1) - 9n}{9}\right]$.

$= \frac{10(10^n - 1) - 9n}{81}$

6. $f(x) = \sqrt{x \sin(x^2)}, \ 0 \leq x \leq \sqrt{\pi}$.

Volume of revolution about x-axis

$$= \pi \int_0^{\sqrt{\pi}/2} x \sin(x^2)\, dx$$

$$= \frac{\pi}{2} \int_0^{\sqrt{\pi}/2} 2x \sin(x^2)\, dx$$

$$= \frac{\pi}{2} \left[-\cos(x^2) \right]_0^{\sqrt{\pi}/2}$$

$$= -\frac{\pi}{2} \left[\cos\left(\frac{\pi}{4}\right) - \cos(0) \right]$$

$$= -\frac{\pi}{2} \left[\frac{\sqrt{2}}{2} - 1 \right]$$

$$= \frac{\pi}{2} \left[\frac{2-\sqrt{2}}{2} \right]$$

$$= \frac{\pi(2-\sqrt{2})}{4}$$

7. Prove by Mathematical Induction

$$\sum_{r=1}^{n} {}^{r}C_1 = {}^{n+1}C_2 \quad , \quad n \in \mathbb{Z}^+.$$

Base Case: $n=1$ $\quad LHS = {}^{1}C_1 = 1$

$RHS = {}^{2}C_2 = 1 = LHS.$

True for $n=1$.

Assume that for some $k \in \mathbb{Z}^+$,

$$\sum_{r=1}^{k} {}^{r}C_1 = {}^{k+1}C_2 .$$

Consider the case $n = k+1$.

$$LHS = \sum_{r=1}^{k+1} {}^{r}C_1 = \sum_{r=1}^{k} {}^{r}C_1 + {}^{k+1}C_1$$

$$= {}^{k+1}C_2 + {}^{k+1}C_1$$

$$= \frac{(k+1)!}{2!\,(k+1-2)!} + \frac{(k+1)!}{1!\,(k+1-1)!}$$

$$= \frac{(k+1)!}{2!\,(k-1)!} + \frac{(k+1)!}{k!}$$

$$= (k+1)! \left[\frac{k+2}{2!\,k!} \right]$$

$$= \frac{(k+2)(k+1)!}{2!\,k!}$$

$$= \frac{(k+2)!}{2!\,k!} = {}^{k+2}C_2$$

Since it is true for $n=1$, and true for $n=k \Rightarrow n=k+1$ is true, by mathematical induction,

$$\sum_{r=1}^{n} {}^{r}C_1 = {}^{n+1}C_2 \,, \quad n \in \mathbb{Z}^+$$

8. (a) Maclaurin series:

(i) $\sin(x^2) = x^2 - \frac{(x^2)^3}{3!} + \dots$

$= x^2 - \frac{x^6}{6} + \dots$

(ii) $\sin^2(x^2) = \left(x^2 - \frac{x^6}{6} + \dots\right)\left(x^2 - \frac{x^6}{6} + \dots\right)$

$= x^4 - \frac{x^8}{6} - \frac{x^8}{6} + \dots$

$= x^4 - \frac{x^8}{3} + \dots$

(b) $4x \sin(x^2) \cos(x^2)$

$= 4x\left(x^2 - \frac{x^6}{6} + \dots\right)\left(1 - \frac{(x^2)^2}{2!} + \dots\right)$

$= 4x\left(x^2 - \frac{x^6}{2} - \frac{x^6}{6} + \dots\right)$

$= 4x\left(x^2 - \frac{2x^6}{3} + \dots\right)$

$= 4x^3 - \frac{8x^7}{3} + \dots$

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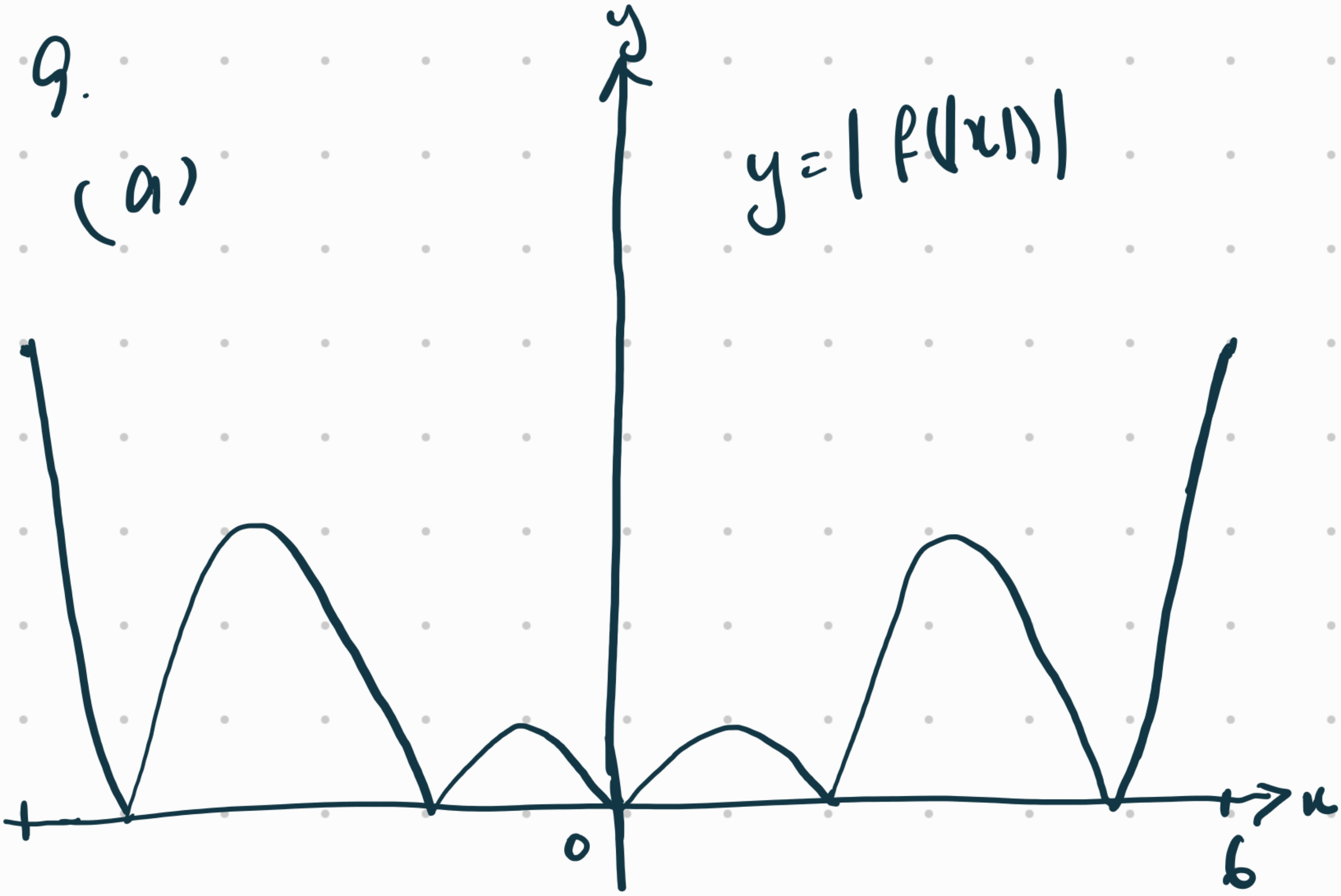
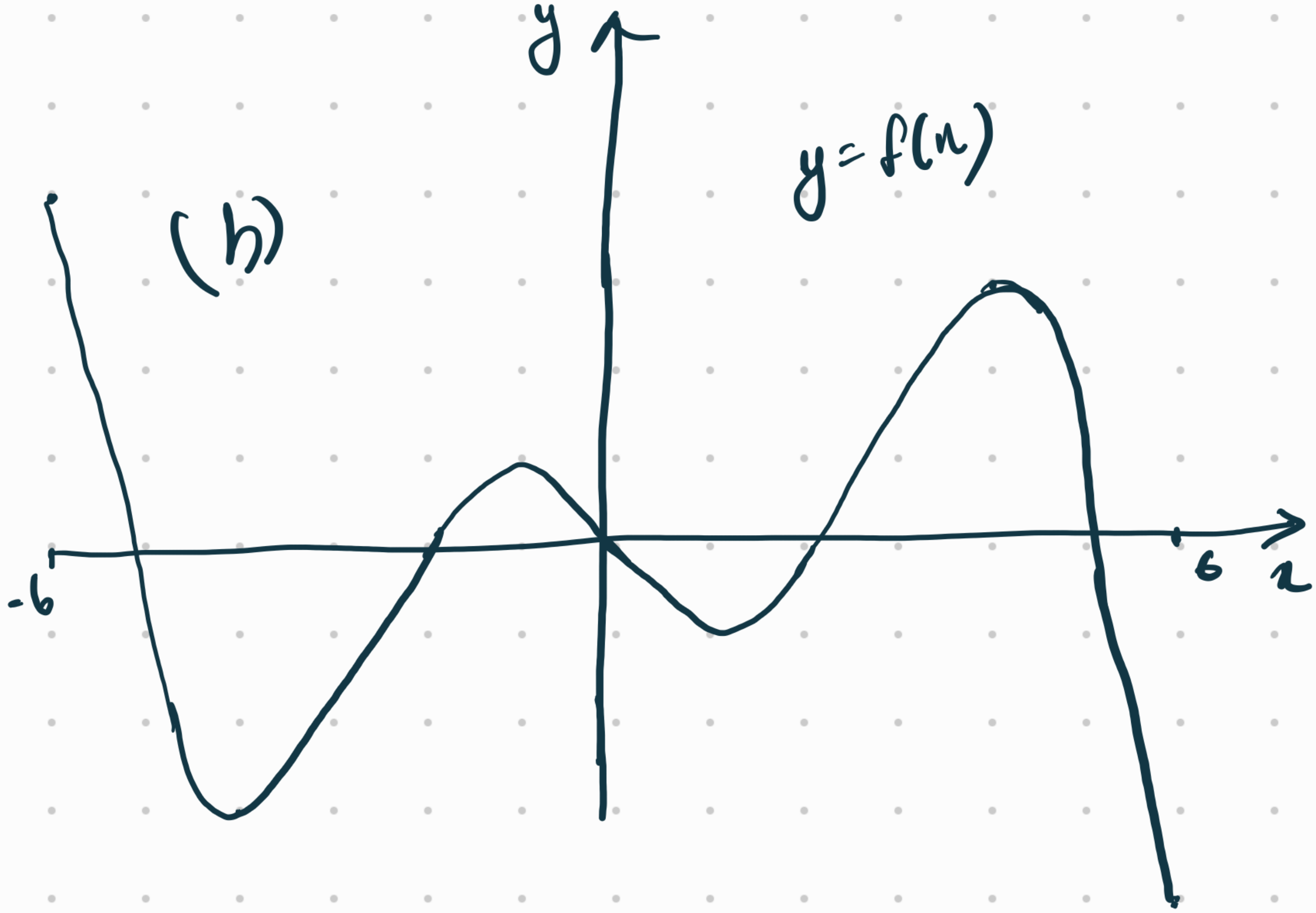
9.

(a)



(b)



(c) Given $\int_0^4 f(|x|)\,dx = 1.6$

(i) $\int_{-4}^{0} f(x)\,dx = -1.6$ (odd $f^{\underline{n}}$)

(ii) $\int_{-4}^{4} f(|x|) + f(x)\,dx$

$= \int_{-4}^{4} f(|x|)\,dx + \int_{-4}^{4} f(x)\,dx$

$= 1.6 \times 2 + 0$

odd function.

$\int_{-4}^{0} f(x)\,dx \quad \int_{0}^{4} f(x)\,dx$

will cancel off.

$= 3.2$

10. $f(x) = \dfrac{4x+2}{x-2}, \ x \neq 2.$

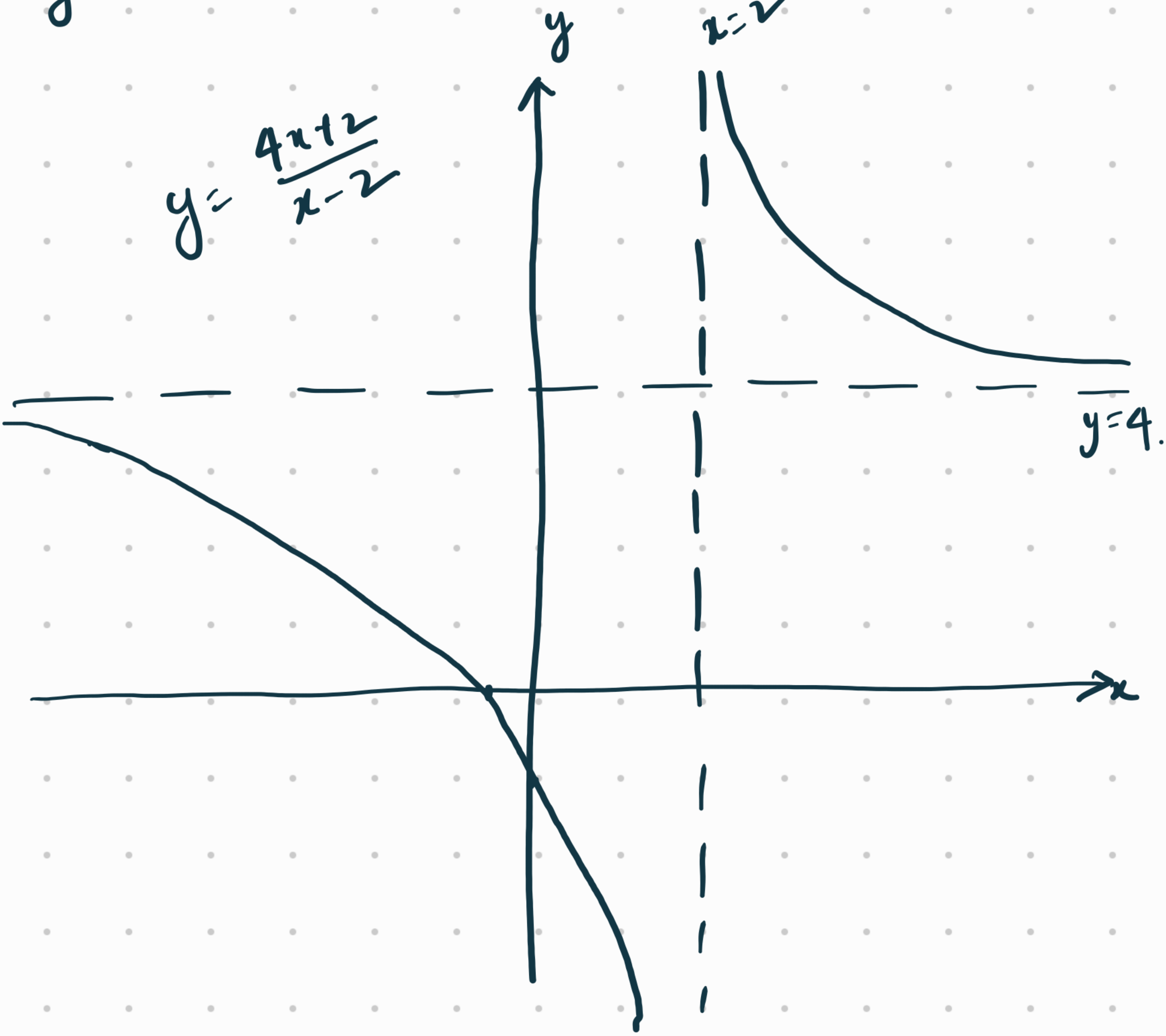
(a) $= 4 + \dfrac{10}{x-2}$

vertical asymptote: $x = 2$.

Horizontal asymptote: $y = 4$.

x intercept: when $y = 0$, $x = -\frac{1}{2}$

y intercept: when $x = 0$, $y = -1$

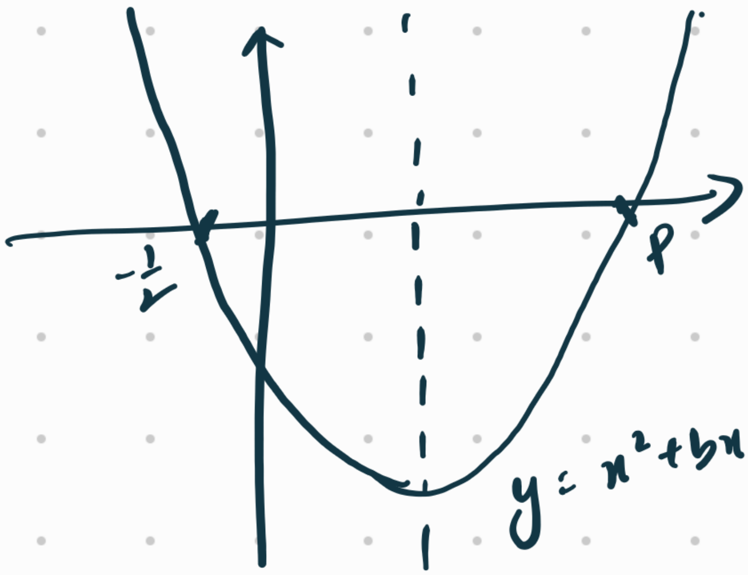


(b) Range of f.

$y \in \mathbb{R},\ y \neq 4.$

(c) $g(x) = x^2 + bx + c$

Axis of symmetry : $x = 2$.



$$\frac{p - \frac{1}{2}}{2} = 2 \Rightarrow p = 4.5 = \frac{9}{2}.$$

(d) $(x + \frac{1}{2})(x - \frac{9}{2}) = 0$

$$x^2 - \frac{9}{2}x + \frac{1}{2}x - \frac{9}{4} = 0$$

$$x^2 - 4x - \frac{9}{4} = 0$$

$$\Rightarrow g(x) = x^2 - 4x - \frac{9}{4}$$

$b=-4, \quad c=-\frac{9}{4}$

(e) $g(2) = 4-4(2)-\frac{9}{4}$

$= -4-\frac{9}{4}$

$= -\frac{25}{4}$

y-coordinate of vertex $= -\frac{25}{4}$.

(f) $f(x) = g(x)$

$\frac{4x+2}{x-2} = x^2-4x-\frac{9}{4}$

$4x+2 = (x^2-4x-\frac{9}{4})(x-2)$

$\Rightarrow x^3-2x^2-4x^2+8x-\frac{9}{4}x+\frac{9}{2}-4x-2=0$

$x^3-6x^2+\frac{7}{4}x+\frac{5}{2}=0$

product of roots $= \frac{-\frac{5}{2}}{1}$

$= -\frac{5}{2}$

11. $P(x) = 3x^3 + 5x^2 + x - 1$

(a) $P(-1) = 3(-1) + 5(1) - 1 - 1$

$= 0$

Hence, $x+1$ is a factor of $P(x)$

(b)

$$P(x) = (x+1)(3x^2+2x-1)$$

$$= (x+1)(3x-1)(x+1)$$

$$= (3x-1)(x+1)^2$$

$$\begin{array}{r|l} & 3x^2+2x-1 \\ \hline x+1 & 3x^3+5x^2+x-1 \\ & 3x^2 \\ \hline & 2x^2+x-1 \\ & 2x^2+2x \\ \hline & -x-1 \\ & -x-1 \\ \hline & 0 \end{array}$$

$$Q(x) = (x+1)(2x+1)$$

(c) $$\frac{1}{Q(x)} = \frac{1}{(x+1)(2x+1)}$$

$$= \frac{A}{x+1} + \frac{B}{2x+1}$$

$$\Rightarrow A(2x+1) + B(x+1) = 1$$

$$2A + B = 0 \quad \text{— (1)}$$

$$A + B = 1 \quad \text{— (2)}$$

$(1) - (2)$: $\quad A = -1$

$\quad B = 2.$

$$\frac{1}{(x+1)(2x+1)} = -\frac{1}{x+1} + \frac{2}{2x+1}$$

(d) $$\frac{1}{(x+1)Q(x)} = \frac{1}{(x+1)^2(2x+1)}$$

$$= \left(\frac{1}{x+1}\right)\left[-\frac{1}{x+1} + \frac{2}{2x+1}\right]$$

$$= -\frac{1}{(x+1)^2} + \frac{2}{(x+1)(2x+1)}$$

$$= -\frac{1}{(x+1)^2} + 2\left[-\frac{1}{x+1} + \frac{2}{2x+1}\right]$$

$$= \frac{4}{2x+1} - \frac{2}{x+1} - \frac{1}{(x+1)^2}$$

(e) $\int \frac{1}{(x+1)^2(2x+1)} dx$

$= \int \frac{4}{2x+1} - \frac{2}{x+1} - \frac{1}{(x+1)^2} dx$

$= 2\ln|2x+1| - 2\ln|x+1| + \frac{1}{x+1} + C$

(f) $f(x) = \frac{P(x)}{(x+1)Q(x)}$, $x \neq -1, x \neq -\frac{1}{2}$

(i) $\lim_{x \to -1} f(x) = \lim_{x \to -1} \left[\frac{(x+1)^2(3x-1)}{(x+1)^2(2x+1)} \right]$

$= \lim_{x \to -1} \left(\frac{3x-1}{2x+1} \right)$

$= \frac{-4}{-1} = 4$

(ii) $\lim_{x \to \infty} f(x) = \lim_{x \to \infty} \left(\frac{3x-1}{2x+1} \right)$

$= \lim_{x \to \infty} \left(\frac{3 - \frac{1}{x}}{2 + \frac{1}{x}} \right)$

$= \frac{3}{2}$

12. $\phi = (a+bi)^3$, $a, b \in \mathbb{R}$.

(a) $(a+bi)^3 = (a+bi)(a+bi)(a+bi)$

$= (a^2 + 2abi - b^2)(a+bi)$

$= a^3 + a^2bi + 2a^2bi - 2ab^2 - b^2a - b^3i$

$= (a^3 - 3ab^2) + (3a^2b - b^3)i$

(i) $Re(\phi) = a^3 - 3ab^2$

(ii) $Im(\phi) = 3a^2b - b^3$.

(b) $(1+\sqrt{3}i)^3$ $\qquad a=1, b=\sqrt{3}$

$= [1-3(3)] + [3\sqrt{3} - (\sqrt{3})^3]i$

$= -8 + (3\sqrt{3} - 3\sqrt{3})i$

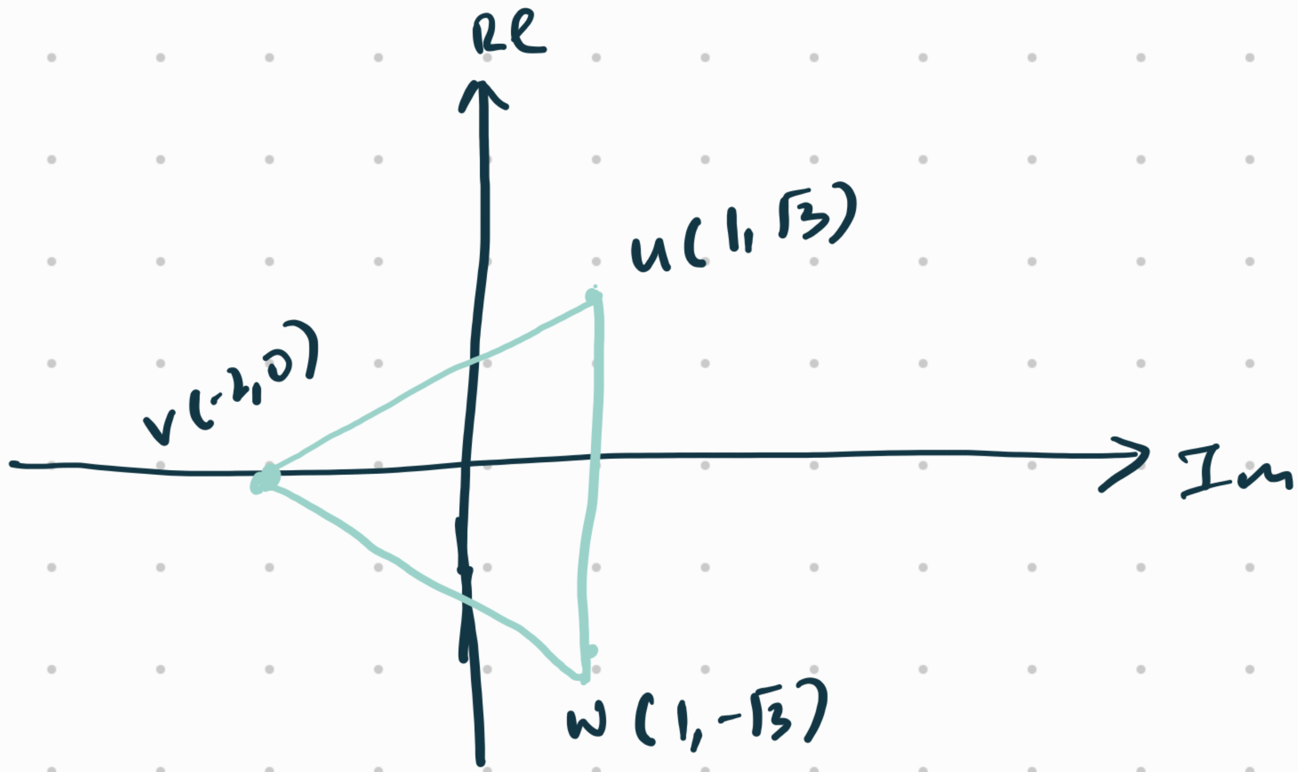
$= -8$.

$z^3 = -8,\quad u = 1+\sqrt{3}i \quad v \in \mathbb{R}.$

(c) $v = -2.$

$w = 1-\sqrt{3}i$ (conjugate)

(d)



Area of $\triangle uvw = \frac{1}{2}(2\sqrt{3})(3)$

$= 3\sqrt{3}$ units^2

(e) $u = 1+\sqrt{3}i = 2e^{i(\frac{\pi}{3})}$

$v = -2 = 2e^{i\pi}$

$w = 1-\sqrt{3}i = 2e^{i(-\frac{\pi}{3})}$

$$u' = 2e^{i\frac{\pi}{3}} \cdot e^{i\frac{\pi}{4}} = 2e^{i\frac{7\pi}{12}}$$

$$v' = 2e^{i\pi} \cdot e^{i\frac{\pi}{4}} = 2e^{i\frac{5\pi}{4}}$$

$$= 2e^{i\left(-\frac{3\pi}{4}\right)}$$

$$w' = 2e^{i\left(-\frac{\pi}{3}\right)} e^{i\frac{\pi}{4}} = 2e^{i\left(-\frac{\pi}{12}\right)}$$

(f) u', v', w' are solutions of

$z^3 = c + di$

$$(u')^3 = \left(2e^{i\frac{7\pi}{12}}\right)^3 = 8e^{i\frac{7\pi}{4}}$$

$$= 8e^{i\left(-\frac{\pi}{4}\right)}$$

$$= 8\left[\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right]$$

$$= 8\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)$$

$$= 4\sqrt{2} - i\left(4\sqrt{2}\right)$$

$c = 4\sqrt{2}$, $d = -4\sqrt{2}$.

(g) $z^n = \alpha$

arguments of u, v, w, u', v', w'

$\frac{\pi}{3}, \pi, -\frac{\pi}{3}, \frac{7\pi}{12}, -\frac{3\pi}{4}, -\frac{\pi}{12}.$

$\Rightarrow \frac{4\pi}{12}, \frac{12\pi}{12}, -\frac{4\pi}{12}, \frac{7\pi}{12}, -\frac{9\pi}{12}, -\frac{\pi}{12}.$

Arranging in order,

$-\frac{9\pi}{12}, -\frac{4\pi}{12}, -\frac{\pi}{12}, \frac{4\pi}{12}, \frac{7\pi}{12}, \frac{12\pi}{12}.$

arguments differ by $\frac{3\pi}{12}$ and $\frac{5\pi}{12}$..

$n = 12$..